GS EXTERIOR EXPERTS · DECK DIVISION · MATERIAL TAKE-OFF + BUILD SET · FOR DECKS & DOCKS

# 281 S Jefferson Ave · Decks + Roof Covers

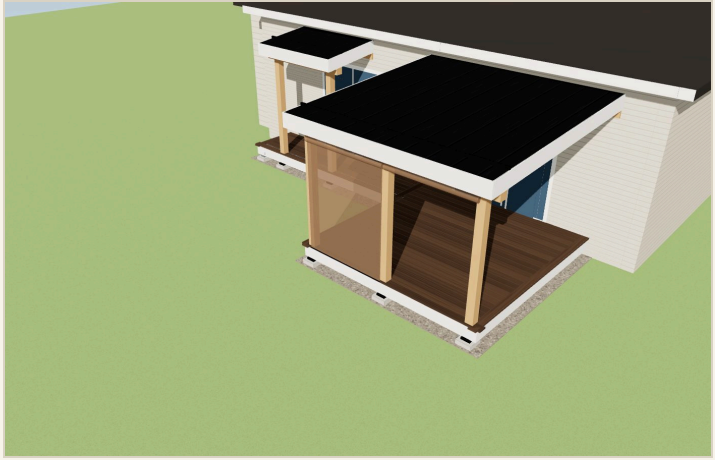
281 S Jefferson Ave, Louisville · two decks (12' x 12' + 14'8" x 3'8" walkway) on CEMCO steel and Diamond Piers · TimberTech Landmark American Walnut, picture frame + dividers, no butt joints · two lean-to covers, black standing seam, Hardie soffit / fascia / rakes · blinds kept · October 16, 2026

Account GSEXT1 · ship to 281 S Jefferson Ave, Louisville, CO 80027 · please quote every line; call Jade Pender 303-550-9558 with substitutions.

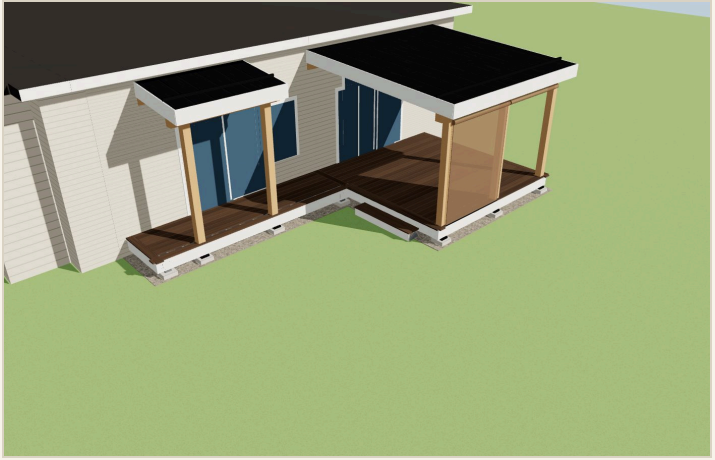
| SQUARE FEET | FOOTINGS | ORDER LINES | HEIGHT |
|---|---|---|---|
| **197.8** | **7** | **78** | **1'-3"** |

## THE JOB

| | |
|---|---|
| Deck | 197.8 SF · 1'-3" above grade · 2 zones, 26'-8" along the house, 12'-0" at the deepest |
| Joists | 7-1/4" CEMCO cold-formed steel C-joists, 16 ga G90, @ 12" OC |
| Rims | single-ply 7-1/4" steel track rims and ledger track on a steel angle |
| Beams | FLUSH boxed 7-1/4" steel C W: 14'-2", 4 posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 3/16" (worst 474 lb); FLUSH boxed 7-1/4" steel C M: 11'-6", 3 posts at 15'-2" / 20'-8" / 26'-2" (worst 1,616 lb) |
| Footings / posts | 2 x 3x3 galvanized steel post stubs + 5 cover posts carrying the deck beam on DP-50/50 |
| Decking | TimberTech Landmark American Walnut — 25 rows parallel to the house, picture frame & dividers in Landmark American Walnut, 3/16" gaps, CONCEALoc |
| Rail | none |
| Stairs | main deck side, toward the walkway: 2 risers @ 7.50", 1 treads, 6 stringers, 5'-0" wide |



Isometric from the stair corner



Isometric from the open corner

**MATERIAL TAKE-OFF** NET = counted off the drawings · ORDER = what to pull and ship

## Lumber

| NET | ORDER | UNIT | ITEM | WHY |
|---|---|---|---|---|
| 347 | **364** | LF | CEMCO 7-1/4" steel C-joist, 16 ga (54 mil) G90 — joists, PF joists, blocking | +5% (cut to length by the supplier where possible) — 12" OC |
| 77 | **82** | LF | CEMCO 7-1/4" steel track, 16 ga G90 — rims (single ply, caps the joist ends) + ledger track | +5% (51 LF rim + 27 LF ledger) |
| 27 | **30** | LF | Steel ledger angle 3" x 3" 14 ga G90 — joist seats on the ledger track | +10% |
| 51 | **54** | LF | CEMCO 7-1/4" steel C, 14 ga (68 mil) G90 — boxed beams (2 C toe-to-toe) | +5% (FLUSH boxed 7-1/4" steel C W 14.2' · FLUSH boxed 7-1/4" steel C M 11.5') |
| 5 | **5** | ea | Steel side bracket, 11 ga galvanized — deck beam bolted to the 6x6 cover post (shared pier) | exact (one per shared post, (2) 1/2" thru-bolts + #12 screws into the beam) |
| 2 | **2** | ea | Steel post stub 3" x 3" x 11 ga HSS, hot-dip galvanized, cap plate welded — cut to height | exact (one per deck-only pier; 6" clear) |
| 2 | **2** | ea | Pier-head post bracket for the Diamond Pier bolt (steel post) | exact |

## Footings

| NET | ORDER | UNIT | ITEM | WHY |
|---|---|---|---|---|
| 7 | **7** | ea | Diamond Pier DP-50/50 (head + 4 x 50" pins, plugs, caps) | exact — confirm stock |
| 1 | **1** | ea | Upgrade to Diamond Pier DP-75/63 at the shared main cover posts (roof + deck on one pier) | exact (20'-8": 3,317 lb vs 3,300 lb) |

## Hardware

| NET | ORDER | UNIT | ITEM | WHY |
|---|---|---|---|---|
| 56 | **62** | ea | Steel clip angle 1-1/2" x 1-1/2" x 7" 16 ga G90 — joist to track / beam | +6 (2 per joist end where it frames into a beam or ledger) |
| 3 | **3** | box | #12-14 x 1" hex-washer self-drilling screw, Climaseal, 500 ct | 1463 needed (clips, track, boxed beams, post brackets) |

| NET | ORDER | UNIT | ITEM | WHY |
|---|---|---|---|---|
| 1 | **1** | box | Structural screw 1/4" x 3-1/2" through the ledger track into the house rim board (or per the engineer) | 40 needed — 2 rows staggered 16" OC |
| 8 | **8** | ea | Lateral-load tension tie for steel joists (Simpson DTT2Z w/ steel-joist kit or the engineer's detail) — IRC R507.9.2 equivalent | exact (2 near each end of every ledger) |
| 2 | **2** | ea | Cold-galvanizing spray (cut ends, field drilling) | lot |

### Flashing & waterproofing

| NET | ORDER | UNIT | ITEM | WHY |
|---|---|---|---|---|
| 1 | **1** | roll | Grace Vycor Deck Protector 12" x 75' (wall membrane behind ledger + side rims) | 27 LF of house wall with laps — nothing on a privacy wall |
| 3 | **3** | ea | Aluminum L-flashing 10' (over ledger + side rims) | house walls only |
| 1 | **1** | roll | 6-mil poly sheeting 10' x 100' — under the decks | 218 SF with laps |
| 1.3 | **1.5** | CY | 3/4" washed gravel, 2" over the poly — under the decks (delivered) | 218 SF x 2" |

### Decking

| NET | ORDER | UNIT | ITEM | WHY |
|---|---|---|---|---|
| 25 | **27** | ea | TimberTech Landmark American Walnut 1x6x12 Grooved | +2 (25 x 12' · 7 x 16' · panel 1 7 rows · panel 2 25 rows · every row one piece) |
| 7 | **9** | ea | TimberTech Landmark American Walnut 1x6x16 Grooved | +2 (25 x 12' · 7 x 16' · panel 1 7 rows · panel 2 25 rows · every row one piece) |
| 4 | **4** | ea | TimberTech Landmark American Walnut 1x6x16 Square Edge — borders & dividers | exact (border 14'-8",12'-0" 12'-1" · border 26'-8",0" 12'-1" · breaker DW/M 12'-1" · border 14'-8",3'-8" 8'-4 15/16") |
| 1 | **2** | ea | TimberTech Landmark American Walnut 1x6x20 Square Edge — borders & dividers | +1 (border 0",3'-8" 14'-9" + border 0",0" 3'-9") |

### Fasteners

| NET | ORDER | UNIT | ITEM | WHY |
|---|---|---|---|---|
| 1 | **1** | box | TimberTech CONCEALoc hidden fastener 500 ct (screws incl.) | 437 needed · 500 supplied |
| 1 | **1** | box | Starborn Cap-Tor xd 2-3/4" Epoxy American Walnut 350 ct | 284 screws needed (first row 28 · border 116 · fascia 92 · stairs 48) |

### Fascia

| NET | ORDER | UNIT | ITEM | WHY |
|---|---|---|---|---|
| 5 | **6** | ea | Deck skirt — HardieTrim 4/4 x 12 x 12' smooth ColorPlus Arctic White (wraps the double rim, in place of the composite fascia) | +1 (left fascia (W) 3'-8 1/2" · right fascia (W) 3'-8 1/2" · front fascia (W) 12'-0" · front fascia (W) 2'-9" · left fascia (M) 12'-0" · left fascia (M) 1/2" · right fascia (M) 12'-0" · right fascia (M) 1/2" · front fascia (M) 12'-0" · front fascia (M) 1") |
| 1 | **1** | box | Color-matched trim screws for HardieTrim, 100 ct (Arctic White) | 40 color-matched screws, pre-drilled, 2 per joist line |

| NET | ORDER | UNIT | ITEM | WHY |
|---|---|---|---|---|
| 1 | **1** | kit | ColorPlus touch-up kit, Arctic White | cut ends |

## Stairs

| NET | ORDER | UNIT | ITEM | WHY |
|---|---|---|---|---|
| 1 | **2** | ea | TimberTech Landmark American Walnut 1x6x12 Square Edge — treads (main deck side, toward the walkway stair) | +1 (1 treads x 2 boards @ 5'-0", 2 per board) |
| 1 | **1** | ea | TimberTech Landmark American Walnut riser board 7-1/4" x 12' (main deck side, toward the walkway stair) | exact (2 risers @ 5'-0") |
| 2 | **2** | ea | Steel stair stringer, galvanized, pre-formed to the riser / tread | exact |

## Porch cover

| NET | ORDER | UNIT | ITEM | WHY |
|---|---|---|---|---|
| 3 | **3** | ea | 6x6x10 Douglas fir #1 post (cover) — Main cover | exact (3 posts on the beam line, field-cut) |
| 2 | **2** | ea | 2x8x16 #1 SYP — cover beam (2 ply) — Main cover | +0 ((2)2x8 x 12.0' at the posts) |
| 1 | **1** | ea | 2x8x16 #1 SYP — cover ledger on the house — Main cover | exact |
| 10 | **11** | ea | 2x8x14 #1 SYP — cover rafters @ 16" OC — Main cover | +1 cull (10 rafters, 13.2' with the overhang) |
| 3 | **4** | ea | 2x8x16 #1 SYP — cover fascia / sub-fascia — Main cover | +1 (38 LF) |
| 6 | **6** | sheet | 7/16" OSB roof sheathing 4x8 — Main cover | 170 SF + 10% cuts |
| 170 | **178** | SF | Standing seam roof — 24 ga steel 16" 1-1/2" mechanically seamed (double-lock, low-slope rated) panels x 13.2' (Black, PVDF), 10 panels — Main cover | 10 panels x 13.2' + 5% (one-piece eave to headwall, no end laps) — mechanical seam priced at the snap-lock rate; confirm with the metal supplier |
| 1 | **1** | roll | High-temp self-adhered ice & water underlayment, 2 sq roll (under every metal panel) — Main cover | 170 SF — full coverage under metal (high-temp) |
| 2 | **3** | ea | Standing seam eave / drip trim 10', color-matched — Main cover | 12 LF low eave + 1 |
| 3 | **4** | ea | Standing seam gable / rake trim 10', color-matched — Main cover | two rakes x 13.2' + 1 |
| 2 | **3** | ea | Headwall (ledger) flashing with Z-closure 10', color-matched — into the house siding / counterflash — Main cover | 12 LF at the house — counterflash into the siding / brick |
| 3 | **3** | ea | Foam + metal closure strips at eave and headwall, 10' — Main cover | eave + headwall closures |
| 1 | **1** | box | Panel clips + pancake-head screws, box of 250 — Main cover | 90 clips @ 18" OC on each seam |
| 1 | **1** | box | Color-matched #10 woodgrip trim screws w/ washers, 250 — Main cover | trims, closures, snow bar |
| 3 | **3** | ea | Butyl tape / polyurethane sealant, per roll or tube — Main cover | headwall, eave, rake laps |
| 12 | **14** | LF | Snow retention — clamp-on bar across the low eave over the deck (Black) — Main cover | 12' — metal sheds 30 psf snow onto the deck and stair below; one bar 12-18" above the eave |
| 5 | **5** | panel | Porch ceiling — HardieSoffit 4' x 10' cedarmill ColorPlus Arctic White (non-vented; 1 in 3 vented if the roof is closed) — Main cover | 170 SF + 10% cuts |

| NET | ORDER | UNIT | ITEM | WHY |
|---|---|---|---|---|
| 3 | **3** | lb | Stainless ring-shank siding nails 2", per lb — Main cover | stainless ring-shank, 6" OC at every rafter |
| 4 | **5** | ea | Cover fascia + rakes — HardieTrim 4/4 x 7-1/4 x 12' rustic ColorPlus Arctic White over the 2x8 sub-fascia — Main cover | +1 (38 LF: front 12' + two rakes 13.2') |
| 1 | **1** | box | Color-matched trim screws for HardieTrim, 100 ct (Timber Bark) — Main cover | color-matched trim screws |
| 3 | **3** | ea | Simpson ABA66Z post base on the shared Diamond Pier bolt — cover post (pier counted with the deck) — Main cover | exact |
| 3 | **3** | ea | Simpson AC6Z post cap, 6x6 to the (2)2x8 beam — Main cover | exact |
| 10 | **10** | ea | Simpson H2.5AZ rafter tie — Main cover | one per rafter at the beam |
| 19 | **19** | ea | FastenMaster LedgerLOK 6" — cover ledger, 2 rows staggered 16" OC — Main cover | 12 LF x 2 rows @ 16" |
| 2 | **2** | ea | 6x6x10 Douglas fir #1 post (cover) — Walkway cover | exact (2 posts on the beam line, field-cut) |
| 2 | **2** | ea | 2x8x16 #1 SYP — cover beam (2 ply) — Walkway cover | +0 ((2)2x8 x 6.3' at the posts) |
| 1 | **1** | ea | 2x8x16 #1 SYP — cover ledger on the house — Walkway cover | exact |
| 5 | **6** | ea | 2x8x8 #1 SYP — cover rafters @ 16" OC — Walkway cover | +1 cull (5 rafters, 4.9' with the overhang) |
| 2 | **3** | ea | 2x8x16 #1 SYP — cover fascia / sub-fascia — Walkway cover | +1 (16 LF) |
| 2 | **2** | sheet | 7/16" OSB roof sheathing 4x8 — Walkway cover | 35 SF + 10% cuts |
| 35 | **37** | SF | Standing seam roof — 24 ga steel 16" 1-1/2" mechanically seamed (double-lock, low-slope rated) panels x 4.9' (Black, PVDF), 5 panels — Walkway cover | 5 panels x 4.9' + 5% (one-piece eave to headwall, no end laps) — mechanical seam priced at the snap-lock rate; confirm with the metal supplier |
| 1 | **1** | roll | High-temp self-adhered ice & water underlayment, 2 sq roll (under every metal panel) — Walkway cover | 35 SF — full coverage under metal (high-temp) |
| 1 | **2** | ea | Standing seam eave / drip trim 10', color-matched — Walkway cover | 6 LF low eave + 1 |
| 1 | **2** | ea | Standing seam gable / rake trim 10', color-matched — Walkway cover | two rakes x 4.9' + 1 |
| 1 | **2** | ea | Headwall (ledger) flashing with Z-closure 10', color-matched — into the house siding / counterflash — Walkway cover | 6 LF at the house — counterflash into the siding / brick |
| 2 | **2** | ea | Foam + metal closure strips at eave and headwall, 10' — Walkway cover | eave + headwall closures |
| 1 | **1** | box | Panel clips + pancake-head screws, box of 250 — Walkway cover | 20 clips @ 18" OC on each seam |
| 1 | **1** | box | Color-matched #10 woodgrip trim screws w/ washers, 250 — Walkway cover | trims, closures, snow bar |
| 3 | **3** | ea | Butyl tape / polyurethane sealant, per roll or tube — Walkway cover | headwall, eave, rake laps |
| 6 | **8** | LF | Snow retention — clamp-on bar across the low eave over the deck (Black) — Walkway cover | 6' — metal sheds 30 psf snow onto the deck and stair below; one bar 12-18" above the eave |

| NET | ORDER | UNIT | ITEM | WHY |
|---|---|---|---|---|
| 1 | **1** | panel | Porch ceiling — HardieSoffit 4' x 10' cedarmill ColorPlus Arctic White (non-vented; 1 in 3 vented if the roof is closed) — Walkway cover | 35 SF + 10% cuts |
| 1 | **1** | lb | Stainless ring-shank siding nails 2", per lb — Walkway cover | stainless ring-shank, 6" OC at every rafter |
| 2 | **3** | ea | Cover fascia + rakes — HardieTrim 4/4 x 7-1/4 x 12' rustic ColorPlus Arctic White over the 2x8 sub-fascia — Walkway cover | +1 (16 LF: front 6' + two rakes 4.9') |
| 1 | **1** | box | Color-matched trim screws for HardieTrim, 100 ct (Timber Bark) — Walkway cover | color-matched trim screws |
| 2 | **2** | ea | Simpson ABA66Z post base on the shared Diamond Pier bolt — cover post (pier counted with the deck) — Walkway cover | exact |
| 2 | **2** | ea | Simpson AC6Z post cap, 6x6 to the (2)2x8 beam — Walkway cover | exact |
| 5 | **5** | ea | Simpson H2.5AZ rafter tie — Walkway cover | one per rafter at the beam |
| 10 | **10** | ea | FastenMaster LedgerLOK 6" — cover ledger, 2 rows staggered 16" OC — Walkway cover | 6 LF x 2 rows @ 16" |

## FASTENER & CONNECTOR SCHEDULE

| | |
|---|---|
| Ledger | 7-1/4" steel ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer confirms) · flashing and membrane as for wood |
| Field | CONCEALoc at every joist in every gap (437) |
| Face screws | Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then alternate top/bottom every 12", pre-drilled · stair treads 2 per stringer per board |
| Steel frame | 7-1/4" C-joists 16 ga @ 12" OC · single-ply track rims · boxed C beams · clip angles with #12 self-drilling screws (4 per leg) · cut ends cold-galvanized · no joist tape needed |

## CUT LIST

| MEMBER | STOCK | LENGTH | QTY | NOTE |
|---|---|---|---|---|
| Field joists (W) | 7-1/4" C | 3'-3 1/2" | 13 | @ 12" OC, crown up |
| Picture-frame joists (W) | 7-1/4" C | 3'-3 1/2" | 2 | centre 5 1/4" from each frame face |
| Side rim plies (W) | 7-1/4" C | 3'-3 1/2" | 4 | 2-ply, laminated on the ground with RSS 2 rows @ 12" staggered |
| Ledger (W) | 7-1/4" C | 14'-8" | 1 | top at deck height less board thickness; membrane behind, flashing over |
| Blocking (W) | 7-1/4" C | 10 1/2" | 14 | 1 row at 1'-10" from the wall |
| Field joists (M) | 7-1/4" C | 11'-7 1/2" | 11 | @ 12" OC, crown up |
| Picture-frame joists (M) | 7-1/4" C | 11'-7 1/2" | 2 | centre 5 1/4" from each frame face |

| MEMBER | STOCK | LENGTH | QTY | NOTE |
|---|---|---|---|---|
| Divider doublers (M) | 7-1/4" C | 11'-7 1/2" | 1 | sister a joist on the divider line so both board edges bear |
| Side rim plies (M) | 7-1/4" C | 11'-7 1/2" | 4 | 2-ply, laminated on the ground with RSS 2 rows @ 12" staggered |
| Ledger (M) | 7-1/4" C | 12'-0" | 1 | top at deck height less board thickness; membrane behind, flashing over |
| Blocking (M) | 7-1/4" C | 10 1/2" | 12 | 1 row at 6'-0" from the wall |
| Field boards panel 1 | Landmark American Walnut | 14'-0" | 6 | one board wall to border, no joints |
| Field boards panel 1 (rip) | Landmark American Walnut | 14'-0" | 1 | one board wall to border, no joints; ripped to fit the border / the house |
| Field boards panel 2 | Landmark American Walnut | 11'-3" | 18 | one board wall to border, no joints |
| Field boards panel 2 | Landmark American Walnut | 11'-4" | 6 | one board wall to border, no joints |
| Field boards panel 2 (rip) | Landmark American Walnut | 11'-4" | 1 | one board wall to border, no joints; ripped to fit the border / the house |
| Stair stringers (main deck side, toward the walkway) | steel stringer | 2'-7" | 6 | 2 risers @ 7.500" · 1 treads @ 11.66" · stringers 12" OC |
| FLUSH boxed 7-1/4" steel C W beam | 7-1/4" C | 14'-2" | 2 | y 3'-6 1/2"; posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 3/16" |
| Steel post stubs / cover posts under FLUSH boxed 7-1/4" steel C W | 3x3 HSS | 6" | 4 | field-measure each |
| FLUSH boxed 7-1/4" steel C M beam | 7-1/4" C | 11'-6" | 2 | y 11'-10 1/2"; posts at 15'-2" / 20'-8" / 26'-2" |
| Steel post stubs / cover posts under FLUSH boxed 7-1/4" steel C M | 3x3 HSS | 6" | 3 | field-measure each |
| Border 26'-8",0" | Landmark American Walnut square | 12'-1" | 1 | mitre at the corners; dividers on the doubled joist |
| Border 14'-8",12'-0" | Landmark American Walnut square | 12'-1" | 1 | mitre at the corners; dividers on the doubled joist |
| Border 14'-8",3'-8" | Landmark American Walnut square | 8'-4 15/16" | 1 | mitre at the corners; dividers on the doubled joist |
| Border 0",3'-8" | Landmark American Walnut square | 14'-9" | 1 | mitre at the corners; dividers on the doubled joist |
| Border 0",0" | Landmark American Walnut square | 3'-9" | 1 | mitre at the corners; dividers on the doubled joist |
| Breaker dw/m | Landmark American Walnut square | 12'-1" | 1 | mitre at the corners; dividers on the doubled joist |
| Stair treads (main deck side, toward the walkway) | Landmark American Walnut square | 5'-0" | 2 | 2 boards per tread, nosing 3/4" over the riser |
| Stair risers (main deck side, toward the walkway) | riser board | 5'-0" | 2 | rip to 7.50" less tread thickness |

| MEMBER | STOCK | LENGTH | QTY | NOTE |
|---|---|---|---|---|
| Left fascia (w) | fascia | 3'-8 1/2" | 1 | flush under the board nose; full board at the front corners, short piece at the house |
| Right fascia (w) | fascia | 3'-8 1/2" | 1 | flush under the board nose; full board at the front corners, short piece at the house |
| Front fascia (w) | fascia | 12'-0" | 1 | flush under the board nose; full board at the front corners, short piece at the house |
| Front fascia (w) | fascia | 2'-9" | 1 | flush under the board nose; full board at the front corners, short piece at the house |
| Left fascia (m) | fascia | 12'-0" | 1 | flush under the board nose; full board at the front corners, short piece at the house |
| Left fascia (m) | fascia | 1/2" | 1 | flush under the board nose; full board at the front corners, short piece at the house |
| Right fascia (m) | fascia | 12'-0" | 1 | flush under the board nose; full board at the front corners, short piece at the house |
| Right fascia (m) | fascia | 1/2" | 1 | flush under the board nose; full board at the front corners, short piece at the house |
| Front fascia (m) | fascia | 12'-0" | 1 | flush under the board nose; full board at the front corners, short piece at the house |
| Front fascia (m) | fascia | 1" | 1 | flush under the board nose; full board at the front corners, short piece at the house |

# G-001 General notes · design criteria

## G-001 GENERAL NOTES · DESIGN CRITERIA

Loads, codes, structure summary, sheet index, site — the permit-set cover

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

GS Exterior Experts · October 16, 2026 · Rev 0

### PROJECT

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel, TimberTech Landmark American Walnut)
281 S Jefferson Ave, Louisville CO 80027
197.8 SF · 1'-3" above grade · 2 zones

### DESIGN CRITERIA

| | |
|---|---|
| **Deck live load** | 40 psf uniform (IRC Table R301.5); guards 200 lb concentrated / 50 plf |
| **Ground snow** | 30 psf (ground; deck design uses the larger of snow and live) — 40 psf live governs; ground snow per the ASCE hazard tool (Louisville criteria) — 30 psf used here (city sheet, non-reducible); Boulder County layer shows 40 psf: engineer confirms |
| **Dead load** | 10 psf deck · 15 psf porch cover (standing seam steel on sheathing, Hardie soffit; 12 psf actual, 15 used) carried on the deck posts |
| **Roof snow (cover)** | 30 psf flat-roof design snow, sliding snow off the house roof + drift at the house wall per ASCE 7 Ch. 7 |
| **Design total** | 50 psf on the deck · cover posts add roof snow + dead to the deck frame and footings (engineered) |
| **Wind** | 145 mph Vult (3-second gust), Exposure C (Louisville 2024 design criteria), Risk Cat. II |
| **Frost depth** | 3'-0" (footings bear below) |
| **Soil bearing** | 1,500 psf presumptive (IRC R401.4.1) — no soils report |
| **Seismic** | B |
| **Wildfire (WUI)** | not in a WUI zone |
| **Codes** | 2024 IRC / IBC as adopted and amended by the City of Louisville (Ord. 1907, Series 2025); IRC R507 decks with LMC 15.05.140; IBC Appendix I patio covers; ASCE 7-22 loads; Louisville Fire Hardening provisions; cold-formed steel frame per AISI S240, engineered |
| **Engineering** | Stamped structural set by a Colorado PE required — the stamped set governs |

### STRUCTURE

| | |
|---|---|
| **Footings** | DP-50/50 |
| **Posts** | 2 x 3x3 galvanized steel post stubs + 5 cover posts carrying the deck beam |
| **Beams** | FLUSH boxed 7-1/4" steel C W: 14'-2", 4 posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 3/16" (worst 474 lb); FLUSH boxed 7-1/4" steel C M: 11'-6", 3 posts at 15'-2" / 20'-8" / 26'-2" (worst 1,616 |
| **Joists** | 7-1/4" CEMCO cold-formed steel C-joists, 16 ga G90, @ 12" OC |
| **Rims** | single-ply 7-1/4" steel track rims and ledger track on a steel angle |
| **Ledger** | 7-1/4" steel ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer confirms) · flashing and membrane as for wood |
| **Decking** | TimberTech Landmark American Walnut — 25 rows parallel to the house, picture frame & dividers in Landmark American Walnut, 3/16" gaps, CONCEALoc |
| **Rail** | none |

### SHEET INDEX


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| A-402 | Roof covers — exploded isometric |
| D-501 | Details |
| D-502 | Connection details — steel frame |
| D-503 | Connection details — roof covers |
| T-400 | Build steps (build set) |
| T-700 | Order + schedules (takeoff) |


### SITE

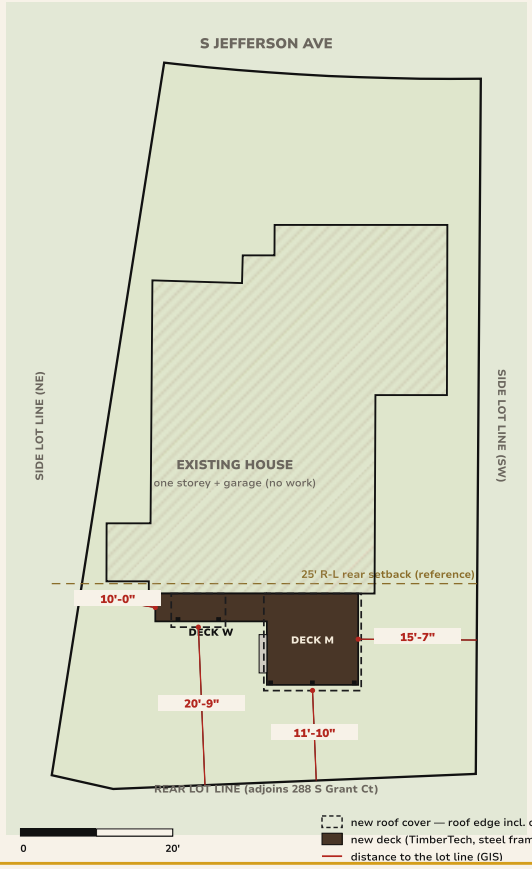
Rear / front lot line: deck edge 12.0' from the line, setback per the PUD...
Side lot line: deck edge 9.8' from the line, setback per the PUD —...

# C-100 SITE PLAN

Parcel, existing house, new decks and roof covers, distances to the lot lines, coverage

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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## PARCEL

281 S Jefferson Ave, Louisville CO 80027
Parcel 157517119033 · Account R0092293
Lot 33, Block 10, Dutch Creek (plat rec. 00492001, 1982)
Section 17, T1S, R69W
Zoning: R-L within the Dutch Creek PUD (approved 12/15/1981)
Lot 4,678 SF · flood zone X
House 1982 one-storey ranch: 920 SF main floor + 400 SF garage (assessor)

## LOT COVERAGE

| | |
|---|---|
| **Lot area** | 4,678 SF |
| **House + garage (assessor footprint)** | 1,320 SF |
| **Existing covers to the posts (2024 aerial, est.)** | ~142 SF (main ~12' x 10'-4", walkway ~5'-4" x 3'-5") |
| **Existing coverage** | ~1,462 SF · 31.3% |
| **Main cover, new (to the posts) 12'-0" x 11'-9"** | 141 SF |
| **Walkway cover, new (to the posts) 5'-4" x 3'-5"** | 18 SF |
| **Proposed coverage** | ~1,479 SF · 31.6% (+~17 SF) |
| **Uncovered decks** | not counted (under 30" above grade) |

## SETBACKS

Decks are 30" or less above grade — not held to the principal setbacks (city deck handout)
Roof covers: part of the principal structure — they replace the two existing covers in place; the 2024 aerial shows the existing main roof ~1'-5" shallower: field-verify the existing post line
R-L reference setbacks (LMC 17.12.040) — the Dutch Creek PUD plan governs; Planning to confirm
Measured (red): to the nearest lot line from the new work — GIS, not survey

## SITE WORK

· Remove the two decks, the two roof covers and the box step; haul off
· Diamond Pier foundations (no excavation spoils), 6-mil poly + 2" gravel under the decks
· Stair lands on grade (no pad)
· Roof drainage: covers shed to grade at the low eave, away from the house
· Call 811 before the piers; protect the lawn and the owner's blinds

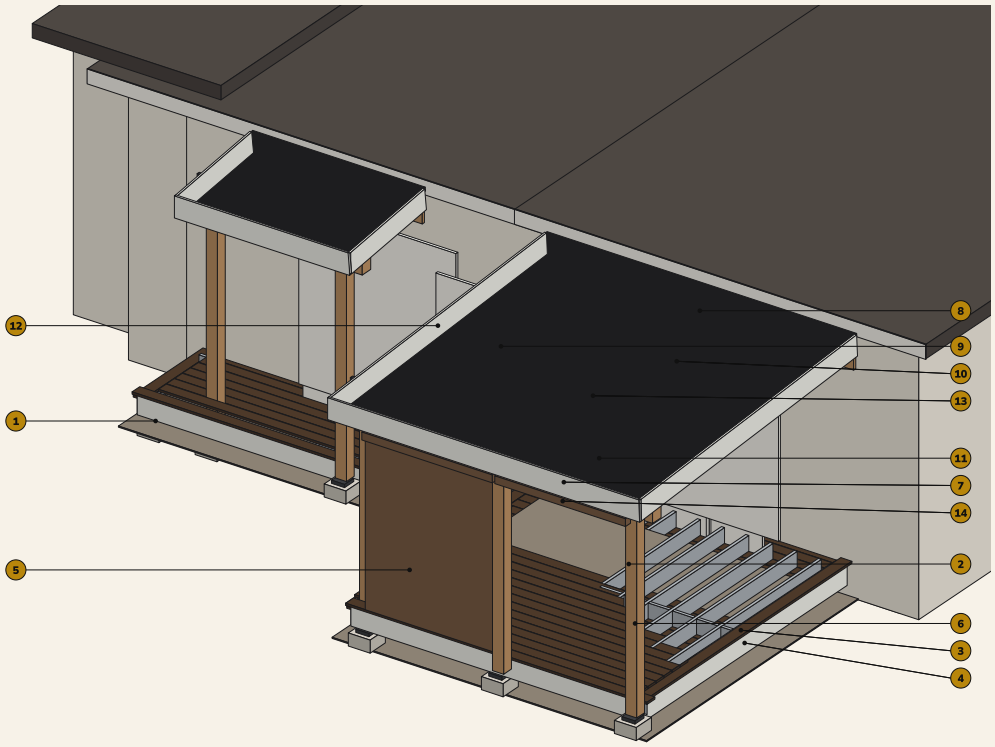
Boulder County GIS parcel + 2024 Pictometry roof outline, Louisville GIS zoning / PUD layers. GIS lines are not a survey — an ILC governs if Planning requires one.

# A-002 ISOMETRIC — ASSEMBLED, LABELED

Everything you see called out: piers, decking, picture frame, skirt, stair, roof covers, finishes

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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**KEY**

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. TimberTech Landmark American Walnut field decking, hidden fasteners
3. Picture frame + breaker boards — TimberTech Landmark American Walnut
4. HardieTrim skirt Arctic White
5. Stair — steel stringers, Landmark treads, closed risers, lands on grade
6. 6x6 Douglas fir #1 cover post on ABA66Z (shared Diamond Pier)
7. (2)2x8 cover beam, wood, AC6Z cap
8. 2x8 cover ledger, wood, LedgerLOK 6" 2 rows, headwall flashed
9. 2x8 rafters @ 16" OC, wood, H2.5AZ at the beam
10. Black standing seam, 24 ga, mechanically seamed
11. Snow retention bar, clamped to the seams
12. HardieTrim fascia + rakes Arctic White
13. HardieSoffit cedarmill ceiling Arctic White
14. Existing roll-down blinds (stay)

House shown for context only (no work). The CEMCO steel deck frame under the decking is called out on S-102 and A-003; cover framing is wood (2x8 rafters, (2)2x8 beams, 6x6 Douglas fir posts).

Axonometric · not to scale · the model the plans are drawn from

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

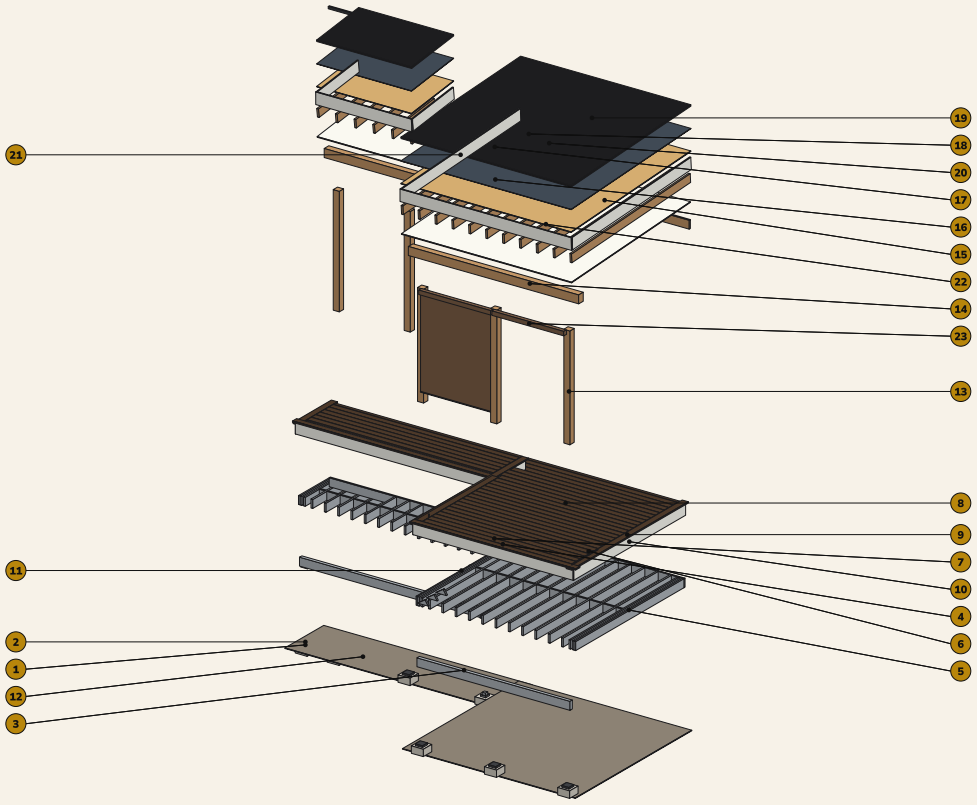
# A-003 Isometric — exploded, labeled

## A-003 ISOMETRIC — EXPLODED, LABELED

Each build layer lifted in order: piers · posts · beams · frame · decking · stair · roof covers layer by layer

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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**KEY**

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 3x3 galvanized HSS post stub on a pier-head bracket
3. Boxed CEMCO 7-1/4" steel C beam (2 C toe-to-toe), flush
4. CEMCO 7-1/4" steel C-joist, 16 ga G90, @ 12" OC — clip angles each end
5. CEMCO 7-1/4" steel track rim (caps the joist ends)
6. Steel ledger track + 3x3 ledger angle on the house rim, flashed
7. Steel blocking at the breaker / picture-frame joists
8. TimberTech Landmark American Walnut field decking, hidden fasteners
9. Picture frame + breaker boards — TimberTech Landmark American Walnut
10. HardieTrim skirt Arctic White
11. Stair — steel stringers, Landmark treads, closed risers, lands on grade
12. Gravel over 6-mil poly under the decks
13. 6x6 Douglas fir #1 cover post on ABA66Z (shared Diamond Pier)
14. (2)2x8 cover beam, wood, AC6Z cap
15. 2x8 cover ledger, wood, LedgerLOK 6" 2 rows, headwall flashed
16. 2x8 rafters @ 16" OC, wood, H2.5AZ at the beam
17. 7/16" OSB roof sheathing
18. High-temp self-adhered underlayment
19. Black standing seam, 24 ga, mechanically seamed
20. Snow retention bar, clamped to the seams
21. HardieTrim fascia + rakes Arctic White
22. HardieSoffit cedarmill ceiling Arctic White
23. Existing roll-down blinds (stay)

Layers lifted in build order. Roof covers pulled apart: posts, beam + ledger, rafters, soffit, sheathing, underlayment, standing seam, snow bar, Hardie trim.

Axonometric · not to scale · the model the plans are drawn from

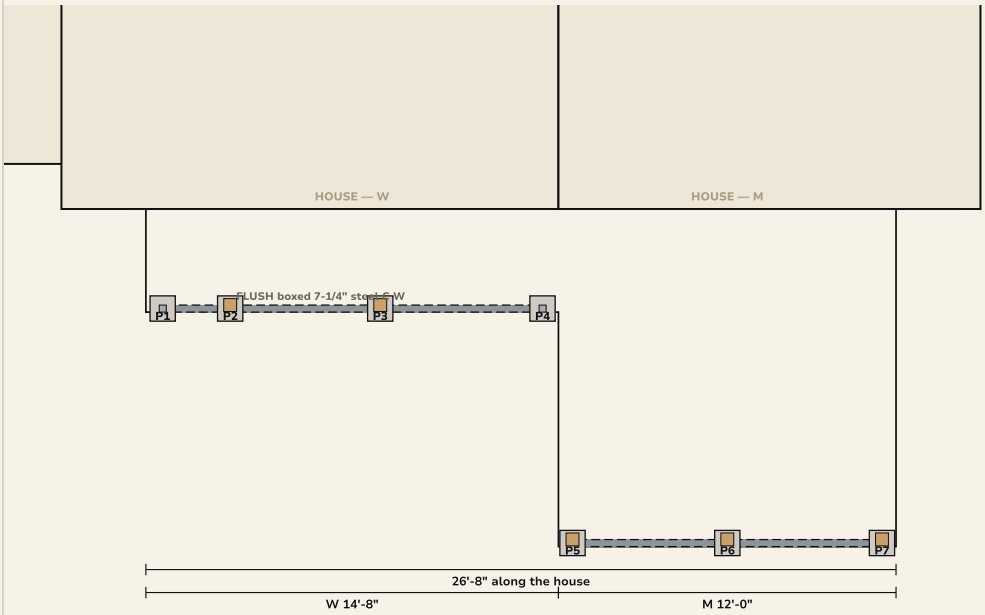
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

# S-100 FOUNDATION PLAN

Footings, posts and beam lines with the footing schedule; beams dashed (below the joists)

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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## FOOTING SCHEDULE

Diamond Pier DP-50/50 (head + 4 x 50" pins), 3,300 lb allowable

| ID | X | Y | ZONE | LOAD |
|---|---|---|---|---|
| P1 | 7 3/16" | 3'-6 1/2" | W | 132 lb |
| P2 | 3'-0" | 3'-6 1/2" | W +cover | 330 lb +roof |
| P3 | 8'-4" | 3'-6 1/2" | W +cover | 474 lb +roof |
| P4 | 14'-1 3/16" | 3'-6 1/2" | W | 273 lb |
| P5 | 15'-2" | 11'-10 1/2" | M +cover | 881 lb +roof |
| P6 | 20'-8" | 11'-10 1/2" | M +cover | 1,616 lb +roof |
| P7 | 26'-2" | 11'-10 1/2" | M +cover | 881 lb +roof |

x from the left frame face facing the house; y from the reference wall.
Frost 3'-0" · soil 1,500 psf · worst post 1,616 lb

Deck-only piers: 3x3 galvanized HSS stub on a pier-head bracket (D-502 / 3). +cover: the 6x6 Douglas fir roof-cover post stands on this pier (ABA66Z) and carries the deck beam on a steel side bracket (D-502 / 4); roof load adds — the engineer sizes the pier.
Beams: boxed CEMCO 7-1/4" steel C, flush (D-502 / 2).

Plan · not to scale · house at the top · dimensions govern

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

# S-101 FRAMING PLAN

Ledger, joists, rims, blocking, beams; every member in the model — the cut list is on T-600

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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## FRAMING NOTES

7-1/4" CEMCO cold-formed steel C-joists, 16 ga G90, @ 12" OC

single-ply 7-1/4" steel track rims and ledger track on a steel angle

7-1/4" steel ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board

blocking one row over the beam, staggered

FLUSH boxed 7-1/4" steel C W: 14'-2", 4 posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 3/16" (worst 474 lb); FLUSH boxed 7-1/4"

Connectors:

Steel clip angle 1-1/2" x 1-1/2" x 7" 16 ga G90 — joist to track / beam: 56

Lateral-load tension tie for steel joists (Simpson DTT2Z w/ steel-joist kit or the engineer's detail) — IRC R507.9.2

Cold-galvanizing spray (cut ends, field drilling): 2

Plan · not to scale · house at the top · dimensions govern

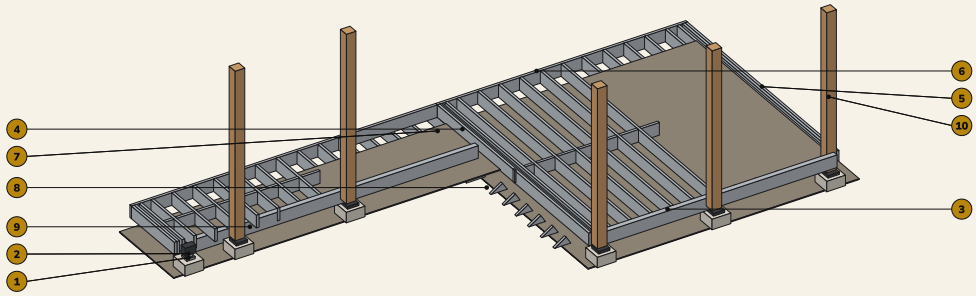
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

# S-102 STRUCTURE ISOMETRIC — STEEL FRAME, LABELED

Piers, stubs, cover posts at the shared piers, boxed beams, C-joists, track rims, ledger track, blocking

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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**KEY**

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 3x3 galvanized HSS post stub on a pier-head bracket
3. Boxed CEMCO 7-1/4" steel C beam (2 C toe-to-toe), flush
4. CEMCO 7-1/4" steel C-joist, 16 ga G90, @ 12" OC — clip angles each end
5. CEMCO 7-1/4" steel track rim (caps the joist ends)
6. Steel ledger track + 3x3 ledger angle on the house rim, flashed
7. Steel blocking at the breaker / picture-frame joists
8. Stair — steel stringers, Landmark treads, closed risers, lands on grade
9. Gravel over 6-mil poly under the decks
10. 6x6 Douglas fir #1 cover post on ABA66Z (shared Diamond Pier)

Decking, covers and house removed to show the frame. Every joist end clips to the track, ledger track or boxed beam; the engineer's clip schedule governs.

Axonometric · not to scale · the model the plans are drawn from

# A-101 Decking plan

**A-101** **DECKING PLAN**

Every board, border and divider as it lays; rail posts shown for the pockets

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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## DECKING

Field: TimberTech Landmark American Walnut, 5.5" x 1", 3/16" gaps

Border & dividers: Landmark American Walnut

Boards run parallel to the house; edge overhang 1" over the fascia

CONCEALoc at every joist in every gap (437)

Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then

Rows per panel: panel 1 (0"–14'-8") 7 rows · panel 2 (14'-8"–26'-8") 25 rows · 2 ripped pieces · 1 breaker — every board one piece wall to border, no butt

Plan · not to scale · house at the top · dimensions govern

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

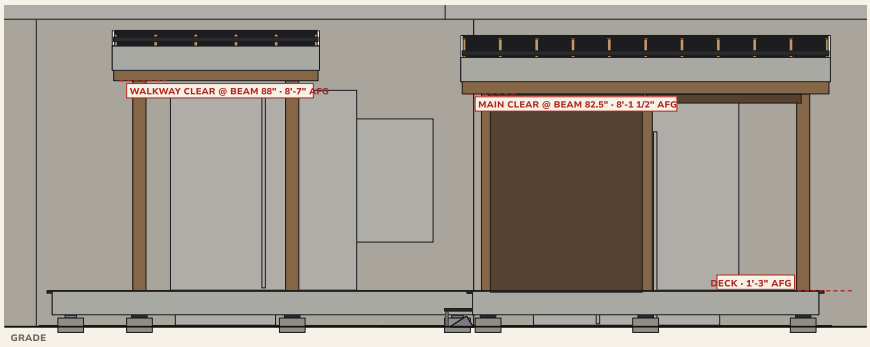
# A-202 ELEVATIONS

Front from the yard and both ends, drawn from the model — deck and roof-cover heights

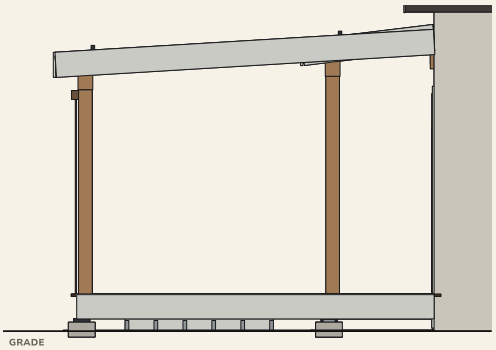
281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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## FRONT ELEVATION — from the yard



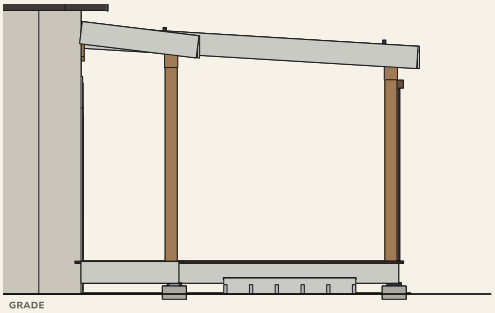

## RIGHT END ELEVATION




## NOTES

- Deck surface 1'-3" above grade (walkway low point 9")
- Heights AFG = above finished grade at the beam line; field-verify grade before setting the piers
- Clear heights are to the underside of the beam / ledger above the deck surface
- Roof covers: black standing seam on 2x8 rafters, Hardie fascia / rakes / soffit; slope to the yard
- Deck frame: CEMCO cold-formed steel; HardieTrim skirt covers the track rims
- No guard required: every walking surface is under 30" above grade (IRC R312.1)

## LEFT END ELEVATION






Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

# A-301 Section + front elevation

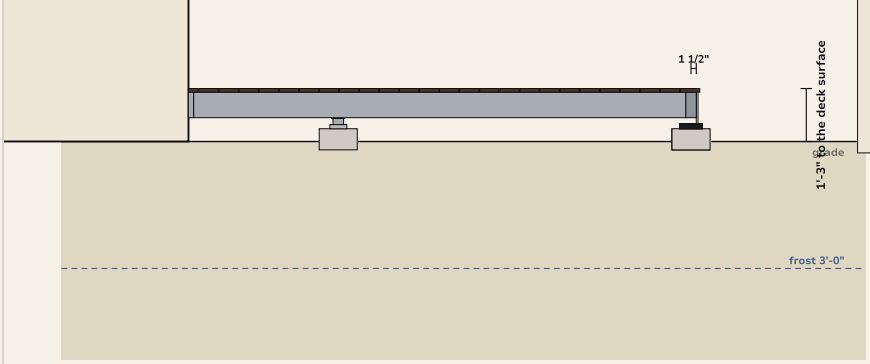
## A-301 SECTION + FRONT ELEVATION

Cut through the deepest zone; heights to grade, frost line, cantilever, guard

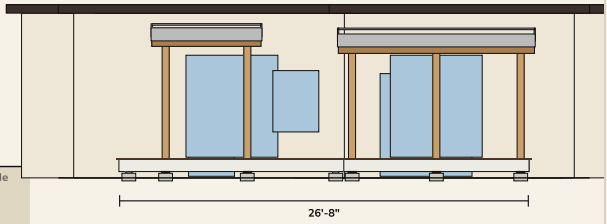
281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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**SECTION through zone M at 6'-0" — looking toward the left end**




**FRONT ELEVATION — from the yard**




### SECTION NOTES

Deck surface 1'-3" above grade

FLUSH boxed 7-1/4" steel C W: 14'-2", 4 posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 3/16" (worst 474 lb); FLUSH boxed 7-1/4" steel C M: 11'-6", 3 posts at 15'-2" / 20'-8" / 26'-2" (worst

Posts 2 x 3x3 galvanized steel post stubs + 5 cover posts carrying the deck beam on DP-50/50

7-1/4" steel ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer confirms) · flashing and membrane as for wood

Section left · elevation right · not to a common scale

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

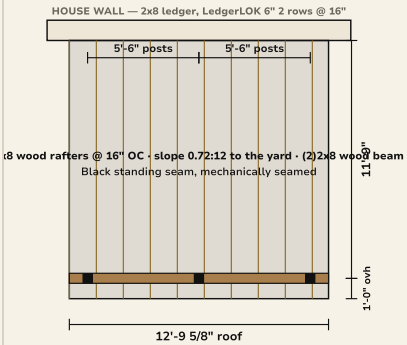
# A-401 ROOF COVERS — PLAN + SECTION

Lean-to covers: ledger, rafters, beam, posts on the piers, clear heights, pitch, standing seam, Hardie, blinds

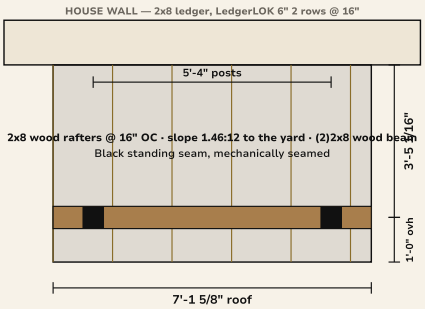
281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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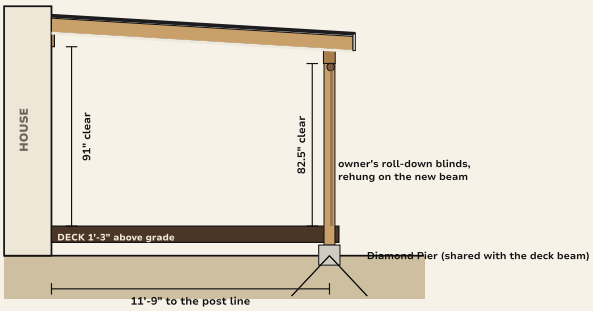
## MAIN COVER — PLAN




## WALKWAY COVER — PLAN

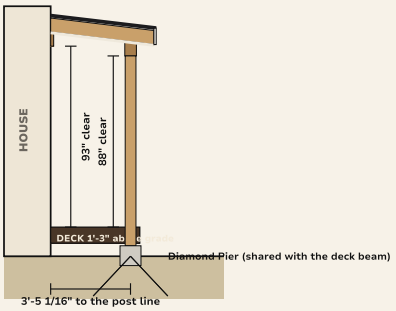



## MAIN COVER — SECTION




Standing seam 24 ga, 16" panels, 1-1/2" mechanical (double-lock) seam — rated for 0.72:12 · high-temp ice & water underlayment over 7/16" OSB
Headwall flashing counterflashed into the siding · color-matched eave / rake trim · snow bar across the low eave
HardieSoffit cedarmill ceiling · HardieTrim fascia + rakes (Arctic White) · H2.5AZ at every rafter · AC6Z caps
6x6 Douglas fir #1 posts on ABA66Z bases on the Diamond Pier bolt · AC6Z cap to the (2)2x8 beam · the engineer's uplift hold-down governs

## WALKWAY COVER — SECTION




Standing seam 24 ga, 16" panels, 1-1/2" mechanical (double-lock) seam — rated for 1.46:12 · high-temp ice & water underlayment over 7/16" OSB
Headwall flashing counterflashed into the siding · color-matched eave / rake trim · snow bar across the low eave
HardieSoffit cedarmill ceiling · HardieTrim fascia + rakes (Arctic White) · H2.5AZ at every rafter · AC6Z caps
6x6 Douglas fir #1 posts on ABA66Z bases on the Diamond Pier bolt · AC6Z cap to the (2)2x8 beam · the engineer's uplift hold-down governs

Plans and sections not to a common scale · dimensions govern · the stamped set governs

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

# A-402 Roof covers — exploded isometric, labeled

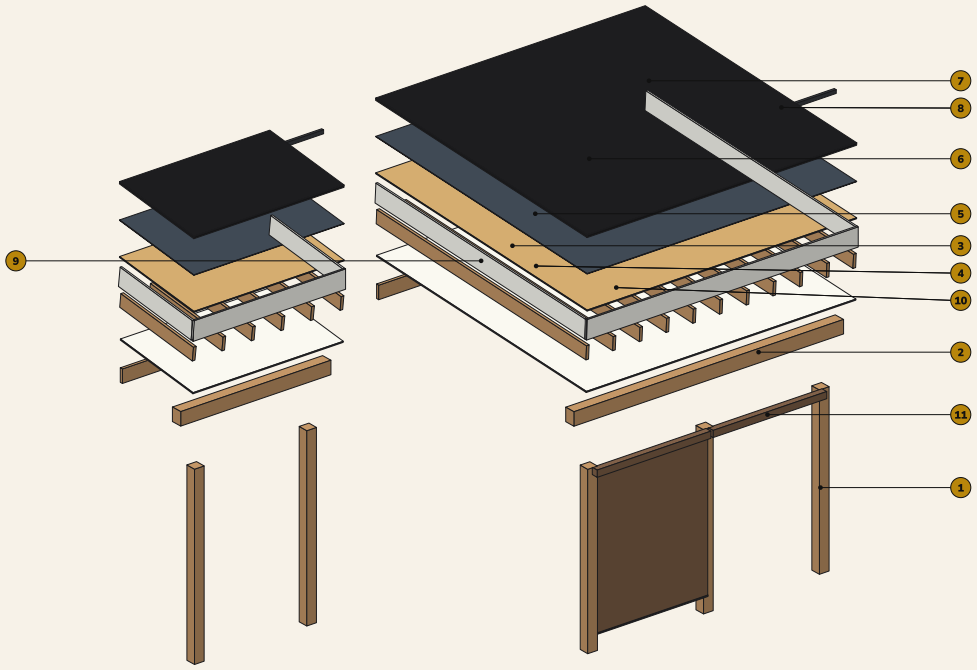
A-402

## ROOF COVERS — EXPLODED ISOMETRIC, LABELED

Both lean-to covers pulled apart layer by layer

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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**KEY**

1. 6x6 Douglas fir #1 cover post on ABA66Z (shared Diamond Pier)
2. (2)2x8 cover beam, wood, AC6Z cap
3. 2x8 cover ledger, wood, LedgerLOK 6" 2 rows, headwall flashed
4. 2x8 rafters @ 16" OC, wood, H2.5AZ at the beam
5. 7/16" OSB roof sheathing
6. High-temp self-adhered underlayment
7. Black standing seam, 24 ga, mechanically seamed
8. Snow retention bar, clamped to the seams
9. HardieTrim fascia + rakes Arctic White
10. HardieSoffit cedarmill ceiling Arctic White
11. Existing roll-down blinds (stay)

Lifts are for clarity only. Order of work: posts on ABA66Z bases, beam on AC6Z caps, ledger on LedgerLOK, rafters with H2.5AZ, soffit, OSB, underlayment, panels, trims, snow bar.

Axonometric · not to scale · the model the plans are drawn from

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

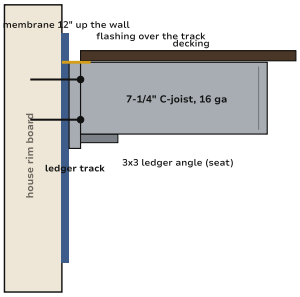
# D-501 DETAILS — STEEL FRAME

Steel ledger · joist clips · boxed beam on the shared pier · Diamond Pier · deck edge · under the deck

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

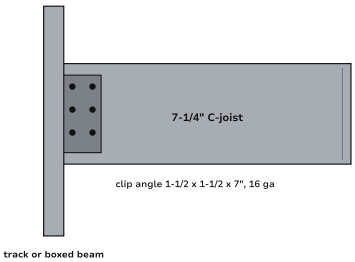
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## 1 STEEL LEDGER AT THE HOUSE



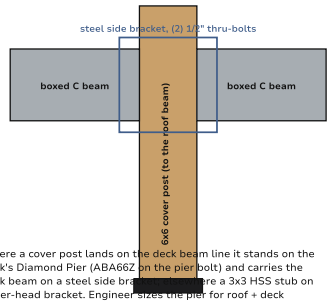

7-1/4" steel ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer confirms) · flashing and membrane as for wood

## 2 JOIST TO TRACK / BEAM



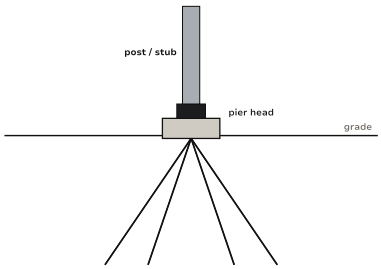

#12-14 self-drilling hex-washer screws, 4 per leg minimum (clip to joist web, clip to track / beam web) — the engineer's clip schedule governs. Cut ends and field holes cold-galvanized.

## 3 BOXED BEAM · SHARED PIER



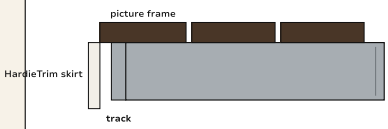

Where a cover post lands on the deck beam line it stands on the deck's Diamond Pier (ABA66Z on the pier bolt) and carries the deck beam on a steel side bracket; there a 3x3 HSS stub on a pier-head bracket. Engineer sizes the pier for roof + deck and the uplift hold-down.

## 4 DIAMOND PIER




DP-50/50 (4 pins) — no excavation; frost-protected by the pins (submit the evaluation report). 811 locate first.

## 5 DECK EDGE · HARDIE SKIRT




TimberTech Landmark American Walnut: hidden fasteners in the field, the picture frame face-fastened with color-matched plugs — every screw the maker's steel-frame screw for 14–18 ga. HardieTrim skirt on furring screwed to the track.

## 6 UNDER THE DECK · LOW END




2" gravel over 6-mil poly, sloped away from the house

Walkway low point: 9" surface to grade leaves 0.75" under the frame — dig out / regrade to 2" min. under that run before the poly.

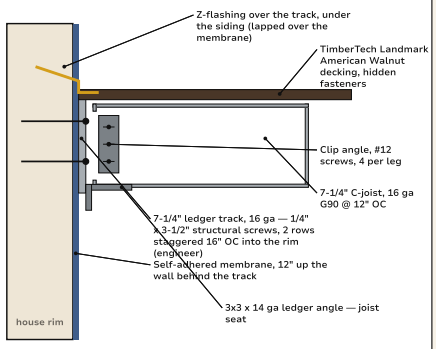
# D-502 CONNECTION DETAILS — CEMCO STEEL DECK FRAME

Ledger track · joist clips · boxed beam · HSS stub on the pier · shared pier · deck edge · stair

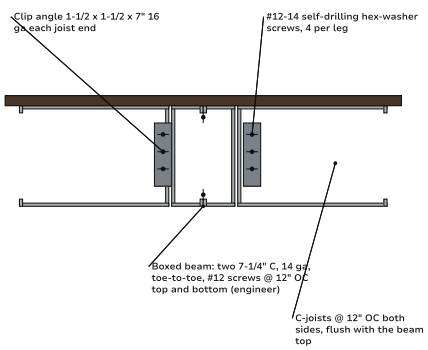
281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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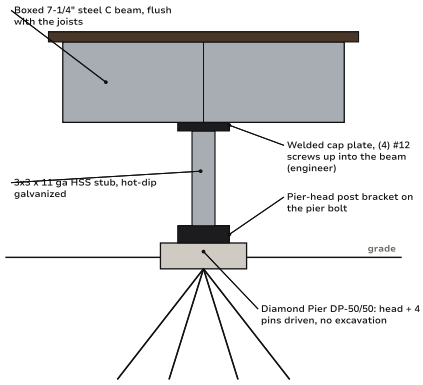
## 1 LEDGER TRACK AT THE HOUSE — SECTION



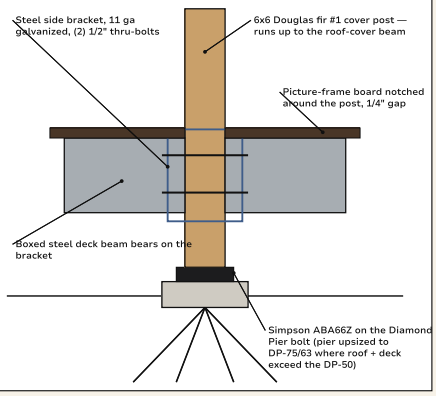

## 2 JOIST TO BOXED BEAM — SECTION



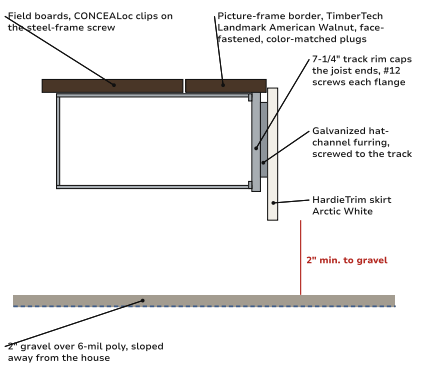

## 3 BEAM ON HSS STUB + DIAMOND PIER



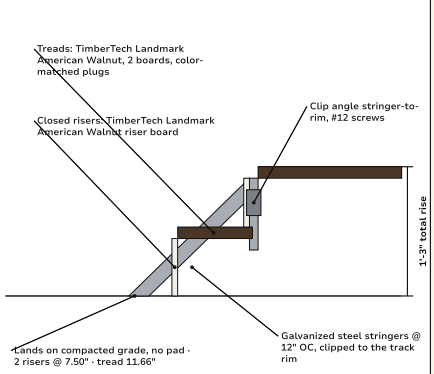

## 4 SHARED PIER — COVER POST + DECK BEAM




## 5 DECK EDGE — TRACK RIM, BORDER, HARDIE SKIRT




## 6 STAIR OFF THE MAIN DECK — SECTION



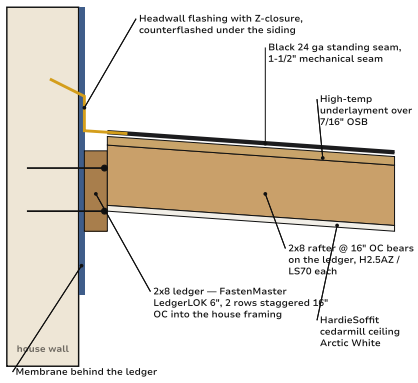
# D-503 CONNECTION DETAILS — ROOF COVERS (WOOD FRAMING)

Ledger + headwall · eave at the beam · post base · rake · standing seam + clip · blinds

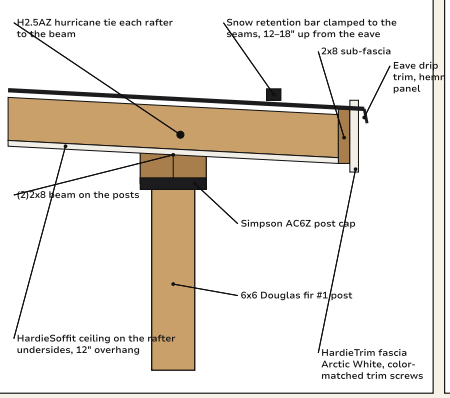
281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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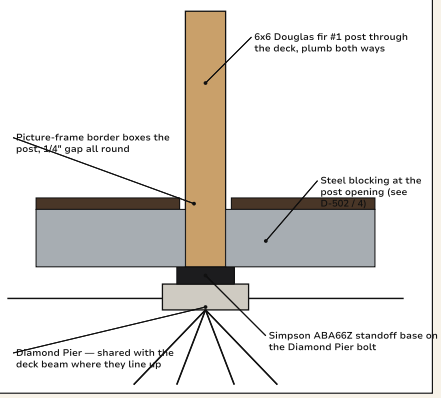
## 1 COVER LEDGER + HEADWALL — SECTION



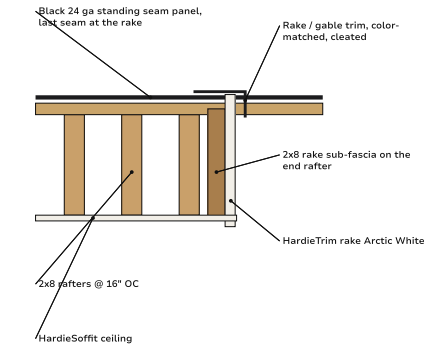

## 2 EAVE AT THE BEAM — SECTION



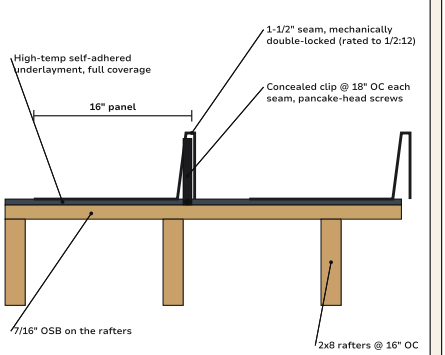

## 3 COVER POST BASE + DECK PENETRATION



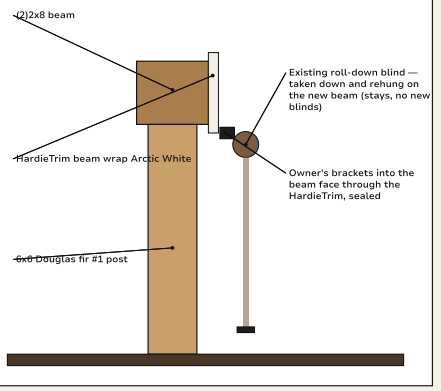

## 4 RAKE EDGE — SECTION ACROSS THE SLOPE



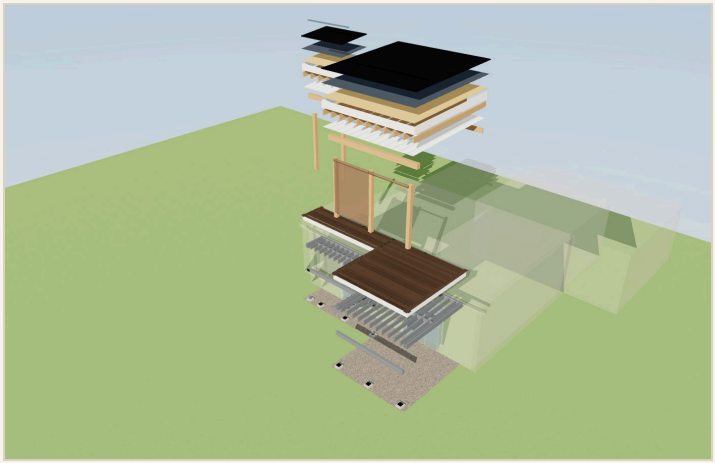

## 5 STANDING SEAM PANEL + CLIP



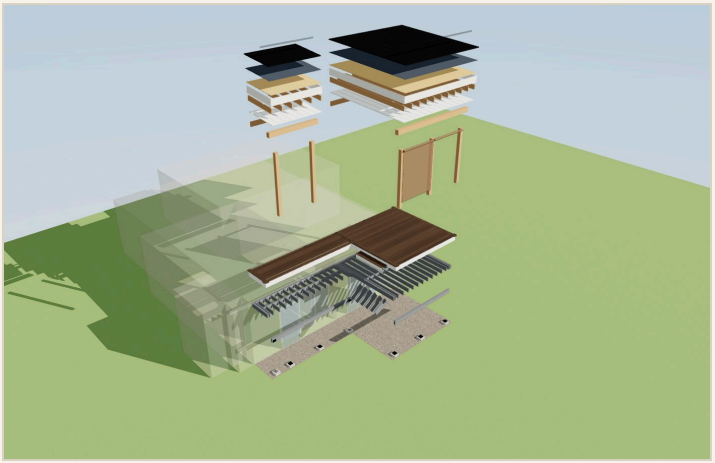

## 6 EXISTING BLINDS AT THE BEAM



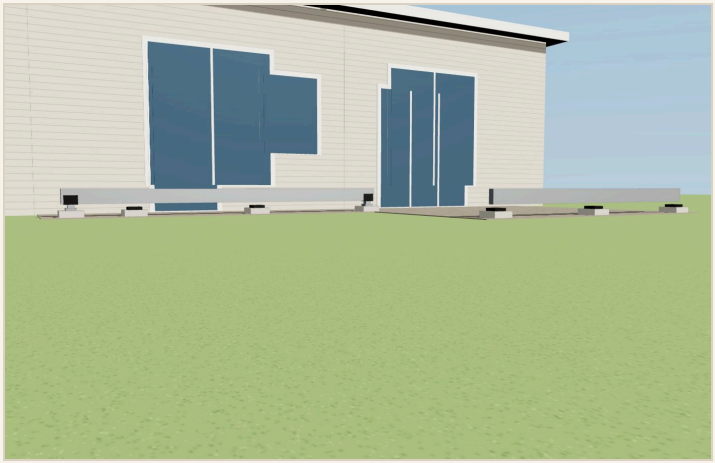
## EXPLODED VIEWS



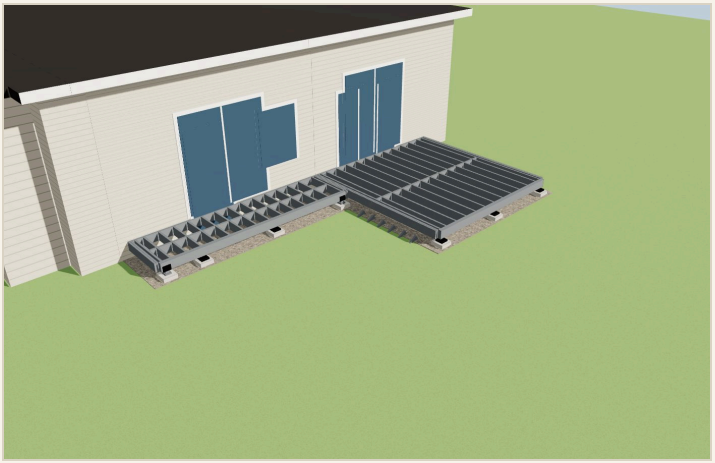
Exploded — every layer



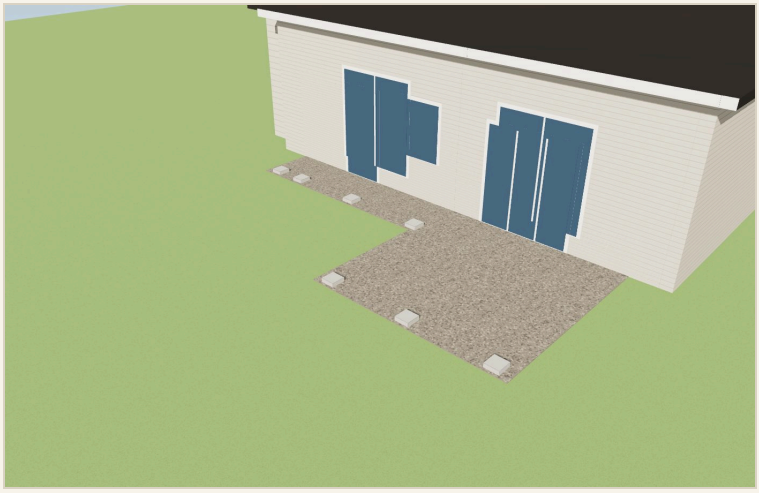
Exploded — from the open corner



Posts and beams on the footings



Frame only — joists, rims, beams, ledger



Step 1 · Footings — 7 Diamond Pier DP-50/50 driven-pin footings

## T-400 · Step 1 · Footings

### GOES IN

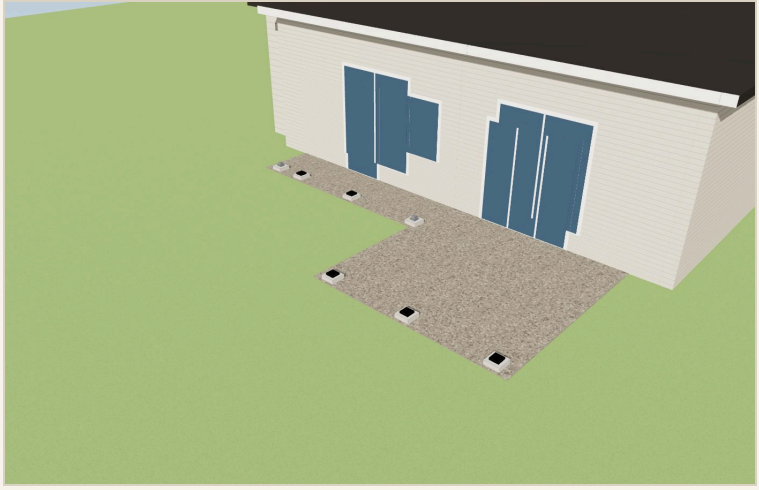
- (7) Diamond Pier DP-50/50 (head + 4 x 50" pins, plugs, caps)
- (1) Upgrade to Diamond Pier DP-75/63 at the shared main cover posts (roof + deck on one pier)

### HOW

- Call 811 and locate before any hole or pin. Strip the existing deck first.
- Pull the beam lines from the house wall: FLUSH boxed 7-1/4" steel C W at 3'-6 1/2" from the reference wall, posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 3/16"; FLUSH boxed 7-1/4" steel C M at 11'-10 1/2" from the reference wall, posts at 15'-2" / 20'-8" / 26'-2".
- Set 7 Diamond Pier heads level at grade on undisturbed soil and drive the 4 pins in each; caps on.

### CHECK

- Footings on the layout within 1/2"; bearing on undisturbed soil
- Pins fully driven, heads level
- Photos before backfill



Step 2 · Posts — 3x3 galvanized steel stubs on pier-head brackets; 6x6 Douglas fir cover posts carry the deck beams where they line up

## T-401 · Step 2 · Posts

### GOES IN

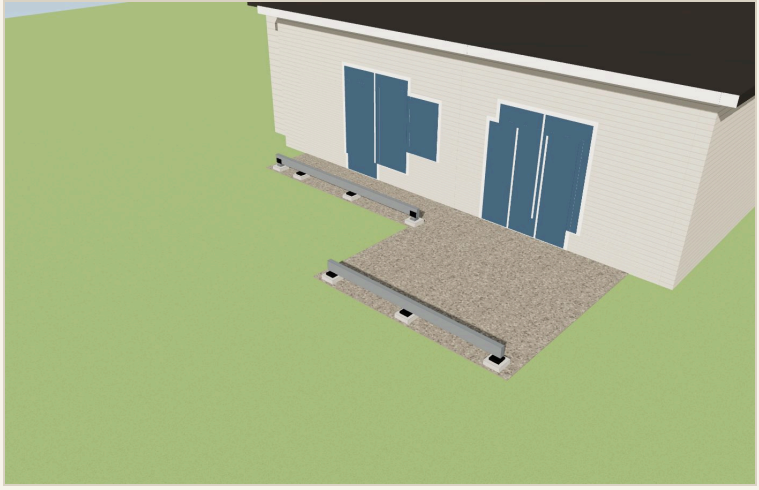
- (5) Steel side bracket, 11 ga galvanized — deck beam bolted to the 6x6 cover post (shared pier)
- (2) Steel post stub 3" x 3" x 11 ga HSS, hot-dip galvanized, cap plate welded — cut to height
- (2) Pier-head post bracket for the Diamond Pier bolt (steel post)
- (3) Simpson ABA66Z post base on the shared Diamond Pier bolt — cover post (pier counted with the deck) — Main cover
- (2) Simpson ABA66Z post base on the shared Diamond Pier bolt — cover post (pier counted with the deck) — Walkway cover

### HOW

- Deck-only piers: 3x3 galvanized HSS stubs on the pier-head brackets, cut so the boxed steel beam top lands at the joist top (flush beams): FLUSH boxed 7-1/4" steel C W 6" / FLUSH boxed 7-1/4" steel C M 6" — field-measure each.
- Where a roof-cover post lands on the deck beam line it stands on the same Diamond Pier on an ABA66Z base; the 6x6 Douglas fir post runs up to the cover beam and the deck beam bolts to it with the steel side bracket.
- Plumb both ways and brace until the beams are on.

### CHECK

- Stubs and cover posts plumb to a string; tops on the beam line
- Every bracket fastened: thru-bolts and #12 screws per the engineer



Step 4 · Beams — FLUSH boxed 7-1/4" steel C W (4 posts) · FLUSH boxed 7-1/4" steel C M (3 posts)

### T-403 · Step 4 · Beams

GOES IN

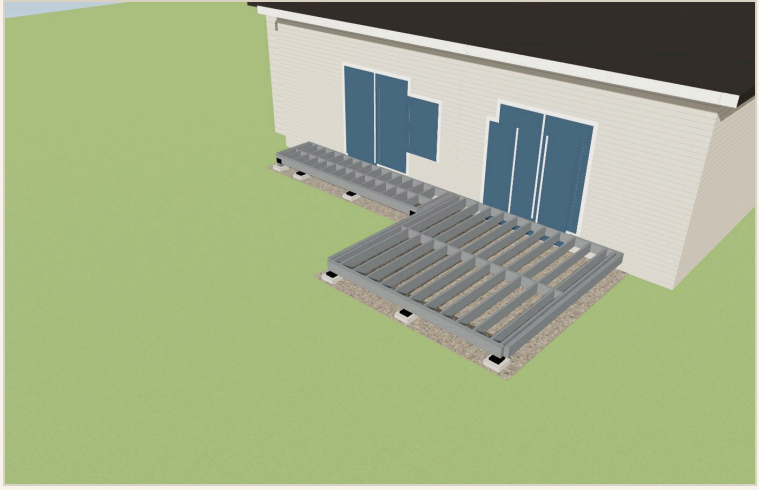
- (51) CEMCO 7-1/4" steel C, 14 ga (68 mil) G90 — boxed beams (2 C toe-to-toe)
- (5) Steel side bracket, 11 ga galvanized — deck beam bolted to the 6x6 cover post (shared pier)

HOW

- FLUSH boxed 7-1/4" steel C W: 14'-2" one piece, 4 posts @ 5'-9 3/16" max
- FLUSH boxed 7-1/4" steel C M: 11'-6" one piece, 3 posts @ 5'-6" max
- Box two C sections toe-to-toe, #12 self-drilling screws per the engineer's pattern; set flush with the joist tops; brackets to the stubs and cover posts.

CHECK

- Beam top = joist top (flush)
- Brackets fully fastened; cut ends cold-galvanized



Step 5 · Frame — CEMCO 7-1/4" steel C-joists at 12" OC, steel track rims, ledger track on a steel angle, clip angles

## T-404 · Step 5 · Frame

### GOES IN

- (347) CEMCO 7-1/4" steel C-joist, 16 ga (54 mil) G90 — joists, PF joists, blocking
- (77) CEMCO 7-1/4" steel track, 16 ga G90 — rims (single ply, caps the joist ends) + ledger track
- (27) Steel ledger angle 3" x 3" 14 ga G90 — joist seats on the ledger track
- (1) Grace Vycor Deck Protector 12" x 75' (wall membrane behind ledger + side rims)
- (3) Aluminum L-flashing 10' (over ledger + side rims)
- (1) 6-mil poly sheeting 10' x 100' — under the decks
- (1.3) 3/4" washed gravel, 2" over the poly — under the decks (delivered)
- (56) Steel clip angle 1-1/2" x 1-1/2" x 7" 16 ga G90 — joist to track / beam
- (3) #12-14 x 1" hex-washer self-drilling screw, Climaseal, 500 ct
- (1) Structural screw 1/4" x 3-1/2" through the ledger track into the house rim board (or per the engineer)
- (8) Lateral-load tension tie for steel joists (Simpson DTT2Z w/ steel-joist kit or the engineer's detail) — IRC R507.9.2 equivalent

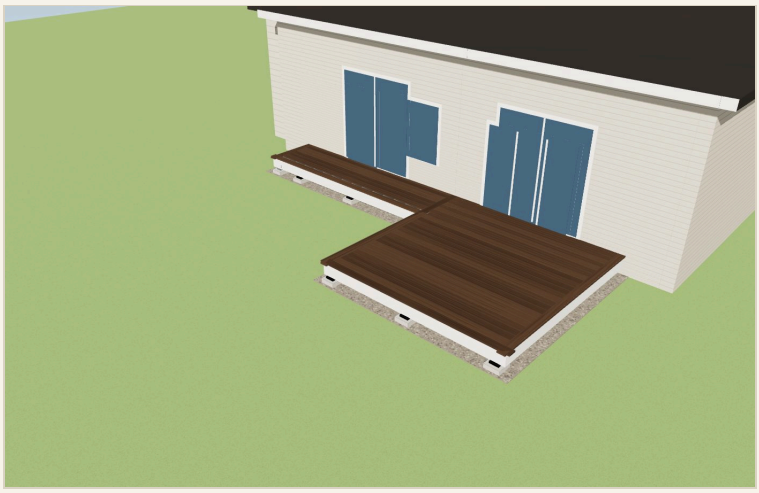
### HOW

- 7-1/4" steel ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer confirms) · flashing and membrane as for wood
- W: 15 CEMCO 7-1/4" C-joists @ 12" OC x 3'-3 1/2"; M: 13 CEMCO 7-1/4" C-joists @ 12" OC x 11'-7 1/2". Joists seat on the ledger angle and clip to the ledger track and the boxed beams with clip angles, 4 #12 screws per leg.
- Track caps every joist end at the rims; picture-frame and breaker joists doubled where the borders land.
- Gravel over 6-mil poly under the frame before the decking (2" min. clear at the walkway low end).
- 7-1/4" C-joists 16 ga @ 12" OC · single-ply track rims · boxed C beams · clip angles with #12 self-drilling screws (4 per leg) · cut ends cold-galvanized · no joist tape needed

### CHECK

- Frame square: diagonals within 1/8"
- Every clip fully screwed; ledger screws per the schedule

- Cut ends cold-galvanized; photos of the ledger, clips and footings before decking



Step 6 · Decking — TimberTech Landmark American Walnut, American Walnut border, hidden fasteners

### T-405 · Step 6 · Decking

GOES IN

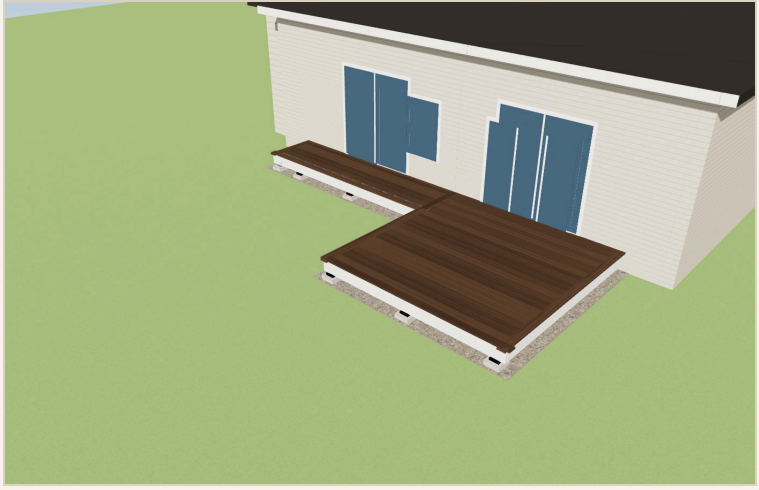
- (25) TimberTech Landmark American Walnut 1x6x12 Grooved
- (7) TimberTech Landmark American Walnut 1x6x16 Grooved
- (4) TimberTech Landmark American Walnut 1x6x16 Square Edge — borders & dividers
- (1) TimberTech Landmark American Walnut 1x6x20 Square Edge — borders & dividers
- (5) Deck skirt — HardieTrim 4/4 x 12 x 12' smooth ColorPlus Arctic White (wraps the double rim, in place of the composite fascia)
- (1) Color-matched trim screws for HardieTrim, 100 ct (Arctic White)
- (1) ColorPlus touch-up kit, Arctic White

HOW

- CONCEALoc at every joist in every gap (437)
- Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then alternate top/bottom every 12", pre-drilled · stair treads 2 per stringer per board
- Rows per panel: panel 1 (0"–14'-8") 7 rows · panel 2 (14'-8"–26'-8") 25 rows · 2 ripped pieces · 1 breaker. Borders and breakers first, then the field rows from the outer edge toward the house.

CHECK

- 3/16" gaps held
- No cut end shows at the outer edge
- Color: every board from the same run



Step 7 · (no rail)

## T-406 · Step 7 · Rail

GOES IN

HOW

CHECK

- —
- Cables tensioned / balusters under 4" clear
- End of construction



Step 8 · Roof covers — 6x6 Douglas fir posts, (2)2x8 beam, 2x8 rafters, OSB, underlayment, standing seam, Hardie soffit / fascia / rakes

## T-407 · Step 8 · Porch cover

### GOES IN

- (3) 6x6x10 Douglas fir #1 post (cover) — Main cover
- (2) 2x8x16 #1 SYP — cover beam (2 ply) — Main cover
- (1) 2x8x16 #1 SYP — cover ledger on the house — Main cover
- (10) 2x8x14 #1 SYP — cover rafters @ 16" OC — Main cover
- (3) 2x8x16 #1 SYP — cover fascia / sub-fascia — Main cover
- (6) 7/16" OSB roof sheathing 4x8 — Main cover
- (170) Standing seam roof — 24 ga steel 16" 1-1/2" mechanically seamed (double-lock, low-slope rated) panels x 13.2' (Black, PVDF), 10 panels — Main cover
- (1) High-temp self-adhered ice & water underlayment, 2 sq roll (under every metal panel) — Main cover
- (2) Standing seam eave / drip trim 10', color-matched — Main cover
- (3) Standing seam gable / rake trim 10', color-matched — Main cover
- (2) Headwall (ledger) flashing with Z-closure 10', color-matched — into the house siding / counterflash — Main cover
- (3) Foam + metal closure strips at eave and headwall, 10' — Main cover
- (1) Panel clips + pancake-head screws, box of 250 — Main cover
- (1) Color-matched #10 woodgrip trim screws w/ washers, 250 — Main cover

### HOW

- Cover ledger on the house at the top of the pitch, LedgerLOK 2 rows; 6x6 Douglas fir posts on ABA66Z bases on the Diamond Piers (shared with the deck beam where they line up), (2)2x8 beam on AC6Z caps; 2x8 rafters @ 16" OC with H2.5AZ ties and the overhang; 2x8 sub-fascia and OSB. Cover framing is wood (SYP rafters / beam, Douglas fir posts).
- High-temp self-adhered underlayment over every sheet, headwall flashing counterflashed into the house wall, eave trim, then the 24 ga snap-lock panels one piece eave to headwall — clips 18" OC on each seam, no face screws in the field; rake trim and closures last; snow retention bar clamped to the seams 12-18" above the low eave, the full width.

- Ceiling: HardieSoffit 4x10 cedarmill Arctic White panels 6" OC stainless ring-shank at every rafter; HardieTrim 4/4 x 7.25 smooth Arctic White fascia and rakes over the 2x8 sub-fascia, color-matched trim screws.

### CHECK

- Ledger flashed and counterflashed — hose test before the ceiling closes
- Posts plumb over blocking, beam level, rafters tied
- Panels straight, seams engaged full length, no oil-canning; snow bar continuous
- Water sheds clear of the deck edge

## 3D MODEL



The interactive 3D model lives at the online version of this document.



Quantities from the drawings; the stamped engineering set governs structure. Quote requested — no pricing on this sheet.